9 November 2026

Dear Editors,

Please consider our manuscript, “Tidal potential measured with distant optical clocks”, for publication as an Article in *Nature*.

Optical clocks connected over long distances can measure changes in gravitational potential. Interpreting these records requires separating overlapping physical contributions and determining the precision with which each can be measured. Tides provide a predictable geophysical signal with which to establish both capabilities and prepare for observations of weaker potential changes.

We measure tidal variations between a strontium clock in coastal Shanghai and an ytterbium clock in inland Wuhan, 692 km apart and connected by a 1,350-km fibre link. The 545.9 hours of effective observations, spanning 353 days, give a response of 0.96 ± 0.11 to the combined solid-Earth and ocean-loading prediction. The lunar semidiurnal group response is 1.00 ± 0.13; both uncertainties are statistical standard errors. Conditional spectra and reciprocal subtraction separately identify the principal lunar and solar-frequency semidiurnal contributions, with each remaining identifiable after controlling for the other. Phase agreement and tests of daily disturbances support the lunar measurement. The solar-frequency contribution remains potentially confounded with laboratory variations at the same frequency.

The weaker constraints on diurnal and separate ocean-loading contributions motivate a quantitative account of measurement capability. We develop a signal-to-noise model that combines signal strength, noise and actual sampling to assess component precision. The model incorporates phase coverage and coupling between fitted components. In its white-frequency-noise implementation, the estimated lunar signal-to-noise ratio is 7.17, close to the measured value of 7.02 (Table 1). Agreement is poorer for the weaker components, which may be more affected by residual contamination through sampling aliases and non-white daily laboratory variations. This comparison tests the precision assessment at the measured amplitudes.

The model connects these results to choices for future observations. Improved clock stability lowers noise, whereas longer continuous records improve phase coverage and reduce component coupling. It also relates precision to effective observing time, conditional on the applicable noise and sampling efficiency. This connection between component separation and measurement capability provides a basis for planning observations of weaker signals after subtraction of the tidal background.

We believe this combination of experiment and measurement assessment will interest Nature’s readers in quantum metrology, gravitation and geophysics. The work establishes how a distant clock comparison can distinguish time-varying potential contributions and assess their precision. Characterizing this tidal background has potential applications to clock networks supporting the redefinition of the second and the unification of height systems at the centimetre level.

The 274-hour summer record is shared with the absolute frequency-ratio study by Hou and colleagues (arXiv:2610.34250), as disclosed in the manuscript. The present work uses the extended record for tidal-component analysis. An independent study has also reported tidal potential variations in a European optical-clock network: K. Stahl et al., *Optical clock network tracks tidal variations of the gravity potential*, arXiv:2610.39635v1 (2026).

Thank you for considering our manuscript. Our suggested referees are listed on the following page.

Yours sincerely,

[Corresponding author name, institution and email to be confirmed]

On behalf of the authors

For your reference, we would like to recommend the following experts as potential referees:

- Jun Ye (jun.ye@colorado.edu) for his pioneering work in optical clocks and frequency ratios.
- Tara M. Fortier (tara.fortier@nist.gov) for her pioneering work in optical clock comparisons and the re-definition of the SI second.
- Sébastien Bize (sebastien.bize@obspm.fr) for his pioneering work in optical frequency standards and the redefinition of the SI second.
- Helen S. Margolis (helen.margolis@npl.co.uk) for her pioneering work in optical frequency standards and clock comparisons.
- Laura C. Sinclair (laura.sinclair@nist.gov), National Institute of Standards and Technology (NIST), Boulder, CO, USA, for her work on free-space optical links and optical clock comparisons.
- M. D. Barrett (phybmd@nus.edu.sg), National University of Singapore, Singapore, for his work on optical frequency standards, systematic uncertainty evaluation and lutetium-ion clock comparisons.